

Helena Teixeira-Dasilva

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Portfolio: <https://teixeiradasilva-helena.github.io/>
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EDUCATION

University of Southern California Masters of Science in Aerospace and Mechanical Engineering - AI and Machine Learning <i>Involvement:</i> GEM Fellowship, Varsity Cross Country and Track, LA Public Schools Tutor <i>Courses:</i> Robotic Dynamics and Control, Mathematical Methods in Deep Learning, Computer Control of Mechanical Systems, Linear Controls II, Robotic Autonomy in Extreme Environments	Aug 2024 - May 2026
Washington University in St. Louis Bachelor of Science in Mechanical Engineering, Second Major Computer Science <i>Honors:</i> Engineering Dean's List (All Semesters); Academic All-American (FL2022, SP2023); President's Council Scholar-Athlete Team Selection <i>Involvement:</i> Varsity Cross Country and Track, BearCubs club - running coach for children with autism, Heat Transfer TA	Aug 2020 - May 2024 GPA: 3.93

EXPERIENCE

Lawrence Livermore National Laboratory <i>Autonomous Systems Engineering Intern</i>	Livermore, CA June 2024 - Aug 2024; May 2025 - present
- Designed, integrated, and flight-tested a UAV autonomy pipeline for precision landing (25–50% accuracy improvement), vision-based collision avoidance, and real-time UAV–boundary intersection prediction using Python, MAVLink, and ArduPilot. - Built a tele-operated UGV system with Arduino-based dual-motor control; delivered a functional test-bench prototype in 2 weeks. - Selected as a top 10 finalist in a lab-wide research symposium among all engineering interns.	
Robot Locomotion And Navigation Dynamics Laboratory <i>Robotics Engineer</i>	Los Angeles, CA Aug 2024 - present
- Developed a ZMP-based quadruped crawling gait in C++ for locomotion while estimating ground stiffness. - Built a human–robot teaming web platform integrating researcher input with Monte Carlo Tree Search for autonomous path planning and field data collection.	
WingXpand™ <i>Robotics Software Engineer Intern (Part-Time)</i>	St. Louis, MO Sept 2023 - Dec 2023
Led simulation efforts and contributed to object detection and hardware testing for an early-stage UAV start-up.	
MIT Lincoln Laboratory <i>Safe Autonomy Research Intern - Integrated Missile Defense Technology</i>	Lexington, MA May 2023 - Aug 2023
- Led full-cycle development of an autonomous UAV landing system using PX4 firmware; achieved successful hardware validation within 6 weeks. - Developed object-oriented software in Python for simple 3D and software-in-the-loop simulations. - Co-led a team project pitching a UUV efficiency-enhancing payload to lab leadership.	
Enterprise Holdings Inc. <i>Software Engineering Intern</i>	St. Louis, MO May 2022 - Aug 2022
- Developed full-stack features for vehicle management applications in Java, JavaScript, HTML/CSS, and SQL. - Designed RESTful APIs and optimized Oracle database workflows.	

SKILLS SUMMARY

Languages	Python, C/C++, Java, MATLAB, SQL, JavaScript, HTML/CSS
Robotics & Tools	ROS, ROS2, ArduPilot, PX4, Gazebo, Simulink, OpenCV, MAVLink, Arduino, Visual SLAM
Dev & Platforms	GIT, Jira, Confluence, SolidWorks, Linux, Windows, LaTeX, NVIDIA Jetson AGX
Soft Skills	Leadership, Communication, Problem Solving, Time Management

References available upon request.